

China's technological catch-up and leapfrogging in electric vehicles: A firm-level study of BYD and CATL

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ABSTRACT

This article examines how Chinese firms BYD and CATL achieved technological catch-up and became global leaders in electric vehicles and EV batteries, disrupting the automotive industry. The global value chain (GVC) and global production network (GPN) approaches struggle to explain how latecomer firms not only join global networks but also transform into lead firms themselves. The article addresses this gap by integrating insights from the economics of innovation literature and putting firm-level technological effort at the centre of analysis. It develops and applies a technological catch-up matrix that captures firm-level effort across five dimensions: proprietary knowledge, organizational capabilities, capital investment, market demand, and local supporting nexus. The article finds that BYD and CATL's success is based on their early investments in lithium-ion battery technology for EVs beginning in the mid-2000s, preceding and actively shaping government industrial policies rather than merely responding to them. Both firms leveraged existing capabilities from China's consumer electronics battery sector while building automotive-specific knowledge and capabilities through strategic partnerships with incumbents and recruitment of foreign expertise. The article's comparative analysis of BYD and CATL with other Chinese EV and EV battery firms demonstrates that while government industrial policies were necessary, firm-level technological effort was equally crucial for achieving market leadership. The article contributes to understanding latecomer industrialization by showing how firms escape subordinated positions in global production networks through systematic technological effort, challenging explanations that attribute China's EV success exclusively to government intervention.

1. Introduction

In 2024, BYD ranked among the top 10 global carmakers by sales for the first time, taking seventh place.² BYD became a global leader in the automotive industry through its move into electric vehicles (EVs), with 4.25 million of the 17.1 million EVs sold globally in 2024, far ahead of Tesla's 1.8 million.³ In this data, EVs refer to pure battery electric vehicles and plug-in hybrid electric vehicles, which is also the definition

used throughout this paper.⁴ While EVs only comprised about one fifth of global annual vehicle sales, the share increased dramatically from 4 % in 2020 to 18 % in 2024.⁵ Technological catch-up is often measured as latercomer firms closing the gap in market share with incumbent firms (Lee & Lim, 2001). By this definition, BYD had caught up with incumbent automakers. Its catch-up occurred largely in the Chinese automotive market – the world's largest automotive market since 2009 and the largest EV market since 2022.⁶ Chinese firms BYD and CATL also

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² Source: Center of Automotive Management, <https://auto-institut.de/presse/marktpositionierung-und-absatzrends-globaler-automobilhersteller/>.

³ Source: Company statements; and IEA for total global EV sales, <https://www.iea.org/reports/global-ev-outlook-2025>.

⁴ A pure EV runs entirely on electricity stored in a large battery. A plug-in hybrid EV, also referred to sometimes as an extended EV or dual-mode, has both a large battery and a gasoline engine; the engine kicks in when the battery is empty, as it was meant to address anxiety about driving range, especially when EV batteries had short ranges. A non-plug-in hybrid EV has a small battery-powered electric motor that is charged by regenerative braking and was created to enhance fuel efficiency using electric assistance; it is fundamentally a gasoline-powered vehicle.

⁵ Source: OICA for total vehicles sales, <https://www.oica.net/category/sales-statistics/>; IEA for EV sales, <https://www.iea.org/reports/global-ev-outlook-2025>.

⁶ In 2024, BYD exported roughly 10 % of its overall vehicles sold (Source: company statement).

overtaken Japanese and South Korean firms to dominate the global EV battery market.⁷

The rise of EV and battery producers in China has set in motion dynamics reshaping the global automotive industry, fundamentally challenging incumbent auto-making countries and forcing responses from established automakers and their governments. This paper examines how Chinese firms captured large market shares in China's automotive market through EVs, radically improving their position in the global automotive industry. Traditional approaches to studying global industries – particularly global value chain (GVC) and global production network (GPN) frameworks – provide important insights into how latecomer firms can participate in global industries. However, these approaches struggle to explain how firms like BYD and CATL not only join global production networks but fundamentally disrupt them by becoming lead firms themselves.

GVC/GPN studies tend to focus on how global industries function, often at a certain point in time, rather than how they change (Grabs & Ponte, 2019). In this context, lead firms' positions are usually treated as given with latecomer firms in a subordinated position. Few studies explicitly acknowledge that for countries to improve their positions in GVCs/GPNs leading to higher value capture, local firms need to break out of subordination and actively improve their position in the GPN hierarchy (Glückler & Panitz, 2016; Hansen et al., 2016). The furthest conceptualization of how this can happen in the GVC and GPN literatures can be found in Yeung (2016) and goes back to the idea of industrial upgrading that emerged in the early GVC literature.

In that early literature, industrial upgrading refers to how local supplier firms from developing countries could improve their position in global value chains by adding products, improving their production process, adding new economic activities (or functions) until they ideally became original brand manufacturers (Gereffi, 1999; Humphrey & Schmitz, 2002). Subsequent contributions using both the GPN and the GVC frameworks critically engaged with the upgrading concept, showing in empirical studies that local firms which achieved product, process and functional upgrading typically did not capture more value (Coe & Yeung, 2015: 172–3; Pickles et al., 2006; Ponte & Ewert, 2009; Tokatli, 2013). The GVC literature then divided into two camps, as some GVC scholars abandoned the concept while others continued to use it without changing. In terms of conceptual development, the GPN literature has confronted the problem more explicitly. Coe and Yeung (2015) emphasize that we need to understand upgrading only in terms of value capture, otherwise it is not upgrading in the economic development sense, and discuss different modes of strategic coupling between supplier firms and lead firms, where some are more likely to lead to value capture than others (see also MacKinnon, 2012).

Yeung (2016) goes furthest in theorizing different modes of strategic coupling and value capture, based on the experiences of East Asian firms. These modes are strategic partnerships with global lead firms, industrial market specialization, and (re)positioning as lead firms. Yeung highlights the importance of three factors which jointly determine whether firms can succeed along any of these three pathways: global competitive dynamics, including windows of opportunity provided by technological and industry-organizational change; domestic political economy conditions, including the industrial policies implemented by the state; and firm-specific initiatives. While the empirical detail is impressive and provides important insights into how various East Asian firms pursued the three pathways, Yeung (2016) does not further develop his conceptual argument.

We argue that to fully understand local firm upgrading, we need to unpack what Yeung (2016: 19) calls firm-specific initiatives of capability building and market development. This is especially the case as firm-specific efforts are the most demanding out of the three factors

(Yeung, 2016: 82). What are the specific initiatives that are required, and how do domestic political economy factors, in particular state industrial policies and existing industrial capabilities, influence whether firms undertake such initiatives?

Existing studies of China's rise in electric vehicles which use the GPN approach discuss how this case differs from traditional coupling cases in that local firm upgrading happened through state-orchestrated market creation and protection, rather than through learning from global lead firms (Schwabe, 2020; Yeung, 2019). While these contributions correctly emphasize the important role of state direction in the Chinese EV case, they cannot fully answer the question posed above, as they lack a conceptualization of how the state policies interact with local firm learning initiatives and vice versa.

To address this gap in the GPN literature and to explain how Chinese firms leapfrogged over incumbent automakers to become lead firms in EVs and EV batteries, we suggest to bring in theoretical tools from the economics of innovation literature, which has long studied firm-specific initiatives under the banner of latecomer technological catch-up. This literature provides a theoretical understanding of technological catch-up from a firm perspective and how government industrial policies and other factors affect firm-level efforts in learning and building capabilities (technological effort, for short). It emphasizes that intra-firm factors such as starting capabilities, knowledge, and ability to create transnational networks for accessing foreign technologies affect their technological effort just as much as government industrial policies. This is important, as most of the existing literature on China's EV transition and leapfrogging by Chinese firms emphasizes the role of government industrial policies but largely overlooks the role of firm-level technological effort.⁸ We build on work arguing that the rise of Chinese EV producers is an example of leveraging windows of opportunity (Altenburg et al., 2022; Whitfield & Wuttke, 2025; Xiong et al., 2022). These works emphasize strong complementarity between the learning and capability building of Chinese firms and government industrial policies, but provide few empirical details on firms' business strategies and capability building processes.

This article fills that gap using extensive empirical data from academic literature by Chinese scholars, industry reports, journalistic accounts, Chinese books about the automotive industry, and company statements to produce an original analysis of China's EV transition. We argue that China's EV industry rise was driven by the early investments and business strategies of BYD and CATL. These two firms moved into lithium-ion batteries (LIBs) for EVs, developing proprietary knowledge and organizational capabilities, before proactive government policies supporting a shift to new energy vehicles emerged. Their effort and consequent success interacted with and shaped government policies from 2009 onwards. Government industrial policies in turn assisted the scaling up and technological upgrading of BYD and CATL, and catalyzed other Chinese battery makers to invest in technological upgrading around EV batteries. Industrial policies also incentivized Chinese automakers to turn to EVs by purchasing batteries from CATL and other Chinese firms. BYD and CATL's central role explains both their dominance as individual firms as well as China's EV industry competitiveness overall. Any explanation of China's rise in EVs must explain why precisely these two firms moved into LIBs for EVs early and how they pursued technological catch-up.

Section 2 presents the technological catch-up approach in the economics of innovation literature and the matrix that we develop based on it. Section 3 summarizes China's automotive catch-up trajectory, showing how Chinese automakers struggled in internal combustion engines and used the EV window of opportunity to capture market share from incumbent automakers. In Section 4, we show that the motivation and success of BYD and CATL with EV batteries stemmed from their

⁷ Source: SNE Research, <https://www.sneresearch.com/en/insight/release-view/371/>.

⁸ This literature includes, for example, de Podestá Gomes et al., 2023, Gong and Hansen (2023), Song et al. (2023), and Thurbon et al. (2023).

Table 1
The five dimensions of technological catch-up.

Proprietary knowledge	Mastering the relevant scientific, technical and tacit knowledge to compete at the technological frontier.
Organizational capabilities	Developing the relevant organizational capabilities to produce and bring to market profitably at scale and achieve the price-performance standards set by competitor firms at the technological frontier.
Capital investment	Accessing the patient capital required to achieve the price-performance standard and to finance losses incurred while learning and scaling up, the timeframe for which is highly uncertain.
Market demand	Accessing sufficient market demand that provides the volumes required for learning and lowering the unit costs of production (i.e. moving down the learning curve), in a context where domestic firms are not yet competitive with incumbent firms.
Local supporting nexus	Creating a local supply chain in components and services, and achieving external economies of scale through industrial clusters, both of which lower firms' production costs.

ability to leverage their knowledge and capabilities from producing batteries for consumer electronics. Section 5 presents our analysis that BYD and CATL not only benefited from government policies but also actively shaped them. Section 6 analyzes the technological catch-up trajectories of BYD and CATL in detail and compares their trajectories to the other top Chinese EV and EV battery producers. We conclude by assessing what the GVC/GPN approaches explain about this case and where these economic geography approaches need supplementing with conceptual tools from the economics of innovation literature.

2. Explaining technological catch-up by latecomer firms

An evolutionary economics view considers technological catch-up as a long run process of closing the gap in knowledge and capabilities by promoting learning and innovation. Furthermore, the technological catch-up process faces uncertainty and continuous change because the pace and nature of technological change within a specific industry is determined by the actions and reactions of incumbent firms and governments as well as latecomer firms and governments (Lee, 2013; Malerba & Lee, 2021). This approach highlights firm-level learning and capability building, and the broader sectoral innovation system in which this occurs.

Malerba (2002, 2004) developed the sectoral innovation system approach to account for how catch-up differs across sectors depending on the configuration of actors and institutions involved. He identified three core elements: the technological regime and knowledge context; the role of other non-firm actors; and the role of national institutions.

National institutions encompass government policies, universities and public research organizations, and financial organizations. More recently, Lee and Malerba (2017) define the sectoral innovation system as consisting of regimes of knowledge and technologies, demand conditions, actors and networks, and institutions that interact to generate a variety of outcomes across sectors and countries in terms of market performance, innovation, growth and industry structure and dynamics. Drawing on the concept of windows of opportunity introduced by Perez and Soete (1988), Lee and Malerba (2017) argue that changes in the technological regime, demand conditions or government policies create discontinuities in sectoral innovation systems. If incumbent firms do not respond to these discontinuities, latecomer firms with relevant accumulated knowledge and capabilities may take advantage of these windows of opportunity. Finally, building on the capability theory of the firm in evolutionary economics, Lee and Malerba (2017) emphasize that all firms share the same sectoral innovation system, but differences in organizational capabilities and learning patterns result in divergent firm performance.

These causal mechanisms constitute the main arguments in the technological catch-up literature as it is used today. We adopt this approach, but seek to extend it by specifying more precisely the causal mechanisms that drive both technological catch-up and leapfrogging during windows of opportunity. The sectoral innovation system perspective itself emerged from the broader national innovation systems literature, which traces back to the work of Christopher Freeman and colleagues in the 1970s on the economics of innovation. However, more recent iterations of the national innovation system perspective, in seeking to create generalizable theories, moved to a level of abstraction that (unintentionally) obfuscates some of the key explanatory factors discussed in the earlier research on the economics of innovation. Freeman's research focused on specific firm strategies that impeded or drove innovation as well as how and why government actions affected the direction and pace of technological change (Freeman, 1974).

Building on this earlier tradition in the economics of innovation, we constructed a technological catch-up matrix that outlines five dimensions along which latecomer firms must close gaps with incumbents (i.e. catch-up) before they can push the technological frontier. This matrix synthesizes insights from Freeman (1974), Perez and Soete (1988), Fransman (1990), and Chandler (2005). As summarized in Table 1, the five technological catch-up dimensions are accessing and mastering proprietary and tacit knowledge; building organizational capabilities to achieve economies of scale; investing fixed capital; accessing the market demand necessary to realize economies of scale; and creating a local supporting nexus that leads to external economies of scale.

All five dimensions are necessary for technological catch-up and

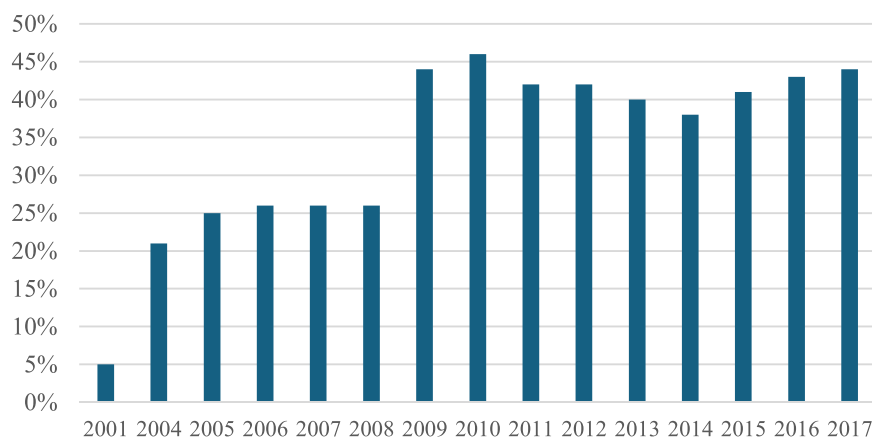


Fig. 1. Share of Indigenous Brand Vehicle Sales in the Chinese Car Market, 2001-2017

Source: Authors' elaboration using data from Feng (2020: 49), based on Chinese government data.

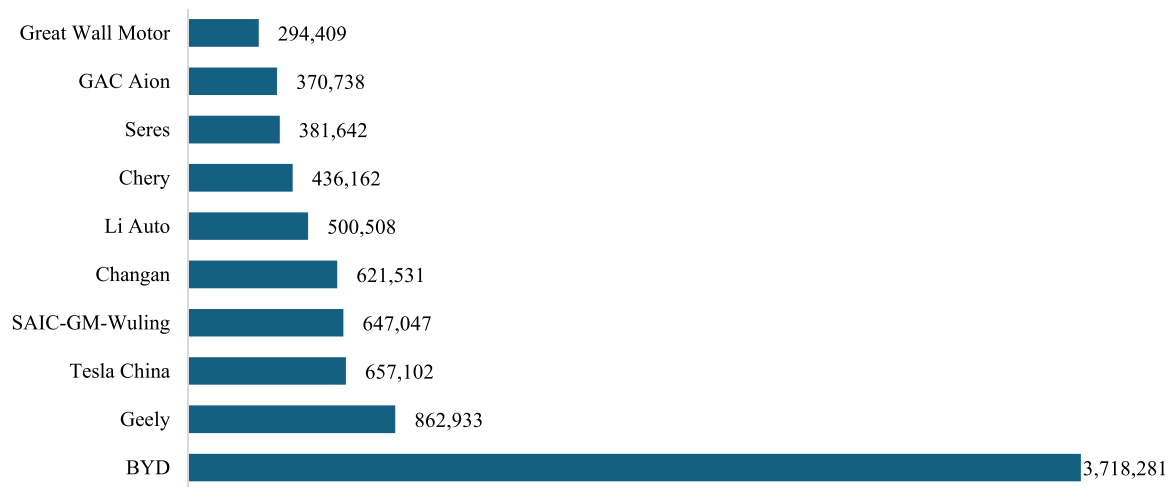


Fig. 2. EV sales volumes in China by carmaker, 2024.

Source: China Passenger Car Association, accessed via CnEVPost, <https://cnevpost.com/2025/01/10/automakers-share-china-nev-market-2024/>.

entail entry barriers for latecomer firms. Latecomer firms must make conscious, purposive and incremental effort to build proprietary knowledge and organizational capabilities by assimilating and adapting existing knowledge (Freeman, 1974; Dahlman & Westphal, 1985; Lall, 1992). But their ability to do so is shaped by the other three dimensions in Table 1. Perez and Soete (1988) were most forceful in arguing that the other three dimensions constituted barriers to entry for latecomer firms because in emerging economies, latecomer firms did not have the same access to capital, market demand and a local supporting nexus that incumbent firms in advanced economies had and which thus gave the incumbents a cost advantage. We take their arguments and adapt them, as latecomer firms (i.e. those firms catching up in a new technology) do not necessarily come from emerging economies but still face constraints related to capital for productive investment, market demand, and a local supporting nexus. In a context where domestic firms are not yet competitive with incumbent firms, they do not have the demand (the sales) required to reach economies of scale and achieve the learning required to go down the learning curve and reduce their costs of production. They also do not have access to the capital required for large-scale investments in production from market-based financial actors because they are not yet profitable and capital is not patient (enough) and risk-taking. And the local supporting nexus is not yet fully developed, increasing the costs of production if inputs and services have to be imported.

It is these dimensions of latecomer catch-up outside the firm, combined with the risky and costly processes of technological effort inside

latecomer firms, that deter firms from pursuing it and from succeeding, which is why state support is required. Technological catch-up often only succeeds if individual firm technological effort is accompanied by government industrial policies, as Perez and Soete (1988) noted, and by similar effort by other firms in the same or adjacent industries to collectively build up the local supporting nexus and create demand through inter-industry linkages, as Chandler (2005) emphasized.

Based on our reading of this literature, we argue that the main objective of industrial policies should be conceived of as reducing the costs and uncertainties that firms face in the technological catch-up process, which occur through four mechanisms: (1) creating initial demand for domestic firms to go down the learning curve; (2) providing low-cost financing, through the financial system or other means such as subsidy programs, which reduces the costs and risks associated with the firm efforts; (3) incentivizing and pressuring firms to invest in technological upgrading; and (4) public investment in infrastructure and industrial policy support for downstream and upstream industries, which help to create the supporting nexus.

Still, government industrial policies cannot substitute for latecomer firms' learning and capability building through technological effort (Dosi & Nelson, 2010). Whether latecomer firms choose to make this effort depends on their perception of their likelihood of success and the return from such an investment. A latecomer firm's perception of what is an adequate rate of return from the investment depends on the firm's existing activities and its need to generate profit, and its ability to acquire and assimilate new technology depends on its existing knowledge and capabilities (Fransman, 1990).

A critical mechanism in catch-up is how latecomer firms access the proprietary and tacit knowledge required, which typically resides in foreign countries and is held by incumbent firms. Proprietary knowledge may be acquired through licensing, FDI, strategic alliances, acquisitions or informal learning (Hobday, 1995). Tacit knowledge, however, is more difficult to access and acquire because it is embedded in practice and context. Thus, partnerships with incumbent firms are often essential (Dosi & Nelson, 2010; Lee & Lim, 2001). Yet as latecomers approach the technological frontier, incumbents may be less willing to collaborate unless it fits with their business strategies (Whitfield, 2023). In both cases, the movement of individuals with tacit knowledge, outside of formal partnerships between incumbent and latecomer firms, can and has served as an important mechanism through which knowledge circulates (Freeman, 1974). Thus, latecomer firms can access foreign knowledge through the movement of individuals, including knowledge at the technological frontier needed for leapfrogging.

These insights about how and why firms are able to build capabilities

Table 2

Vehicle sales (all types) by brand in China, 2017 versus 2024.

2017		2024	
Total vehicle sales	23,582,833	Total vehicle sales	22,893,993
Top 10 brands	16,704,356	Top 10 brands	16,152,887
1. VW	4187,340	1. BYD	3718,281
2. GM	2867,662	2. VW	2808,578
3. Nissan	1525,449	3. Geely	1773,251
4. Honda	1440,139	4. Toyota	1568,234
5. Geely	1382,658	5. Changan	1365,745
6. Toyota	1313,899	6. Chery	1335,205
7. Hyundai	1198,011	7. SAIC	1298,496
8. Ford	1135,582	8. Honda	881,366
9. Great Wall	1001,129	9. BMW	714,530
10. SAIC	652,587	10. GAC	689,201
Local brands/top 10	18 %	Local brands/top 10	63 %

Note: Chinese brands in *italic*.

Source: Centre of Automotive Management, <https://auto-institut.de/e-mobility-2/>.

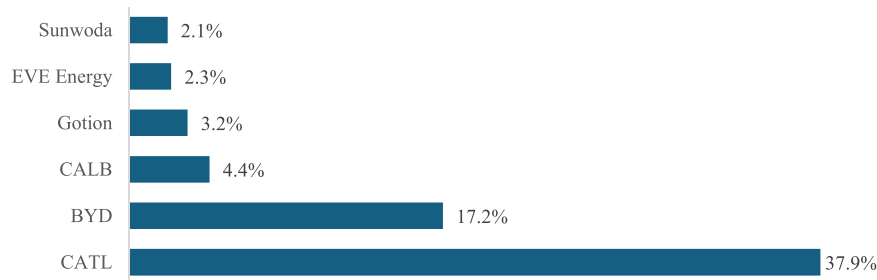


Fig. 3. Chinese EV battery firms' global market share, 2024, based on annual cumulative global battery usage for xEV (GWh).

Source: SNE Research, https://www.sneresearch.com/en/insight/release_view/371/.

and the respective role of government policies inform the analyses in Sections 4, 5 and 6. Specifically, we apply the technological catch-up matrix to examine how BYD and CATL became global leaders as well as why they outperformed other EV and EV battery firms in China (see Tables 3 and 4 in Section 6). The key factor was that they built partnerships with incumbent firms and leveraged foreign knowledge from the technological frontier early, which allowed them to learn and innovate in EV battery technology and production processes.

3. Incumbent advantage in the automotive industry and China's catch-up through EVs

The central Chinese government sought to create an automotive industry through the policy of 'trading market access for technology'. This policy required that foreign incumbent automakers that wanted to operate in China had to form joint ventures with state-owned enterprises (SOEs) since the 1980s, and the government limited the number of local firms in the automotive sector through a licensing system. The 'trading market access for technology' approach did not lead to the indigenization of technology in the joint ventures, but the significant foreign investments in China's automotive industry did build local supply chains and the capabilities of Chinese engineers. The skilled labor and local supply chains in turn were essential to the emergence of independent Chinese automakers such as Chery, Geely and Great Wall Motor in the late 1990s.⁹ These independent Chinese automakers initially offered lower-cost imitations of the foreign models but gradually indigenized technology and offered local models and brands (Chu, 2011; Li, 2009; Feng, 2010, 2016, 2020). Independent Chinese automakers gained significant shares in the Chinese market during the 2000s but then stagnated from 2010 onwards (Fig. 1). This is because the foreign incumbent automakers in joint ventures with Chinese SOEs responded to the new competition by lowering their prices and upgrading their local production facilities and supply chains, and thus continued to outcompete Chinese automakers in the middle- and high-end product segment (Brandt & Thun, 2016; Feng, 2016).

Why was it so difficult for independent Chinese automakers to catch up in internal combustion engine (ICE) technology? The answer is that incumbent advantages of the first mover firms in the US and the fast followers in Europe and Japan remained strong because automotive production is characterized by high levels of tacit knowledge and no disruptive technological innovations. While tacit knowledge is high in every industry, it constituted a particular catch-up challenge in the

automobile industry for several reasons. ICE technology is less science-based than other industries, as it relies primarily on mechanical engineering where knowledge and capabilities are tacit and cumulative (see Abernathy, 1978; Colias, 2025; MacDuffie, 2013). Additionally, the automobile is an assembly of parts, but they must be designed in an integrated manner to ensure safety and optimal performance, which makes tacit knowledge very important as system design is embedded in a firm's organizational routines (Cabigiosu et al., 2013; Henderson & Clark, 1990; MacDuffie, 2013; Zirpoli & Becker, 2011). The dominant design of ICE vehicles has remained unchanged in its core fundamentals since Ford commercialized their mass production in the early 20th century, so tacit knowledge has accumulated in incumbent automakers for almost a century (Bergek et al., 2013).

Lee and Lim (2001) argue that catch-up is less likely to happen in sectors where technology is more tacit, but that innovations which disrupt the technological trajectory make catching up relatively easier because it reduces incumbents' advantages related to tacit knowledge in the dominant technology (see also Lee, 2013). The South Korean experience of technological catch-up in automobiles proves these points. South Korean firms were the only latecomers in the second half of the 20th century to catch-up with incumbent automakers, and Hyundai was the only Korean firm to retain its position in the top ten automakers globally, in terms of revenue. Hyundai was unique among the Korean firms in the extent of its technological efforts to indigenize knowledge related to the internal combustion engine and capabilities related to systems integration as well as marketing and distribution.¹⁰ Hyundai achieved indigenization through partnerships with foreign technology firms and individuals who had worked in incumbent automotive firms and who had very good knowledge of export markets, especially the US, as well as through aggressive learning efforts via reverse engineering and sending their engineers for overseas training. Independent Chinese automakers such as Geely and Chery, and also BYD in ICE cars since 2003, used similar strategies as Hyundai.¹¹

Alternative technologies to the internal combustion engine have existed for some time, especially fuel-cell based EVs and battery EVs. US legislation in the early 1990s spurred incumbent automakers to consider alternative technologies, but battery technology was not sufficiently developed at the time (i.e. firms used lead-acid batteries with short range instead of LIBs), and the incumbents did not dedicate significant R&D resources to their EV projects but rather sought to appease emissions regulators with the least amount of change (Dyerson & Pilkington, 2005). Toyota made the most progress, developing the Prius, which, however, was a non-plug-in hybrid, closer in design to an ICE car than an EV. Other Japanese incumbents started building EVs with LIBs based on improved chemistries and convinced the Japanese government in 2010

⁹ We use the term independent Chinese automakers to distinguish them from the foreign-SOE JVs. We cannot call them 'private' firms, as many of them had local government shareholders or origins in smaller SOEs where the shareholding changed over time. The key point is that they did not rely on the technology of foreign incumbents via JVs and emerged independently (and initially often illegally) from the central government's automotive industry strategy, which controlled entry into the market and only started to give production licenses to non-SOEs in the early 2000s.

¹⁰ For a synthesis of the South Korea automotive industry story in general and Hyundai in particular, see Whitfield and Wuttke (2024).

¹¹ On Chery and Geely, see Luo (2005), Li (2009), Chu (2011), Feng (2010, 2016, 2020), Feng and Li (2019), and also the synthesis in Whitfield and Wuttke (2024).

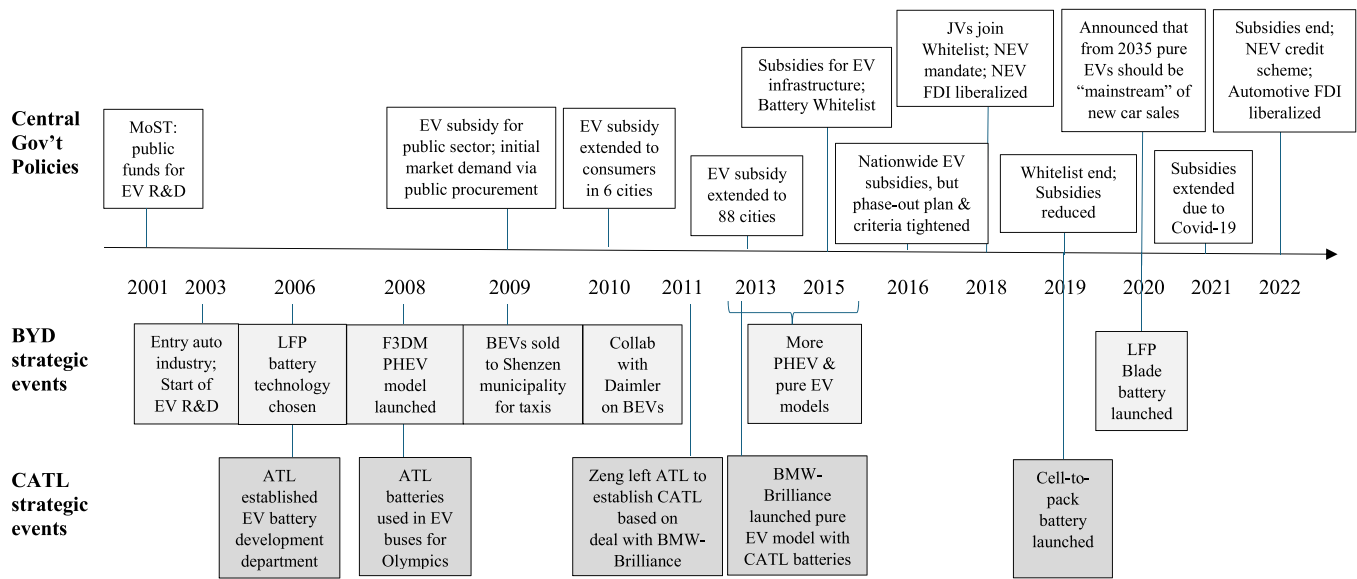


Fig. 4. Timeline of BYD and CATL strategic events and Chinese central government policies.

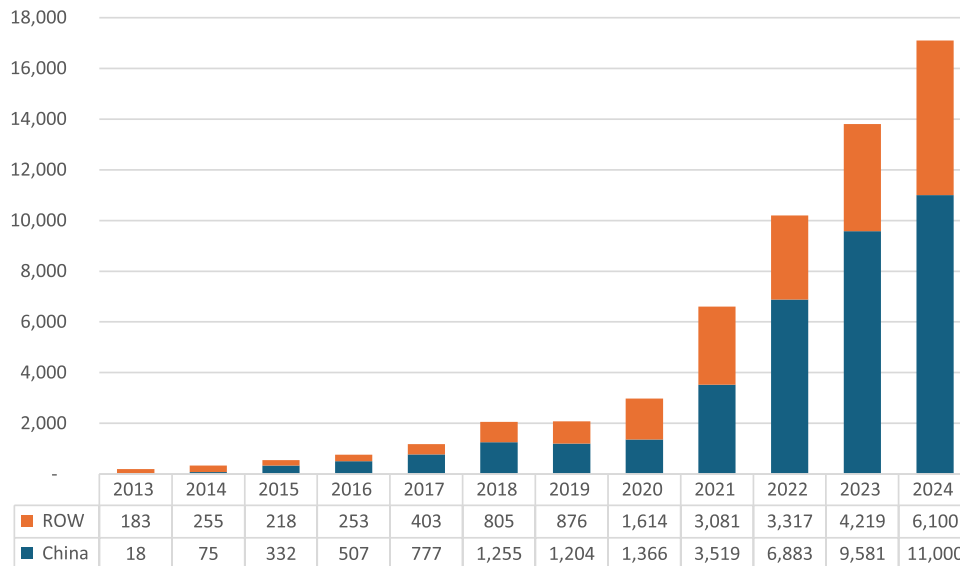


Fig. 5. China and rest of the world (ROW) EV sales in '000 units, 2013–2024.

Source: Authors' elaboration based on data from IEA for global EV sales, <https://www.iea.org/reports/global-ev-outlook-2025>; and China Association of Automobile Manufacturers for Chinese EV sales, accessed via <https://www.statista.com/statistics/425466/china-annual-new-energy-vehicle-sales-by-type/>.

to provide subsidies for consumer purchases, but then the government shifted in 2012 to supporting hydrogen-based fuel cell technology as the future of the automobile industry, which South Korean automaker Hyundai also followed (Tilleman, 2015; Thurbon et al., 2023). German automakers actively resisted an early move to battery EVs (Meckling & Nahm, 2018). US automaker GM undertook serious efforts to develop EVs in the 1990s with lead-acid batteries, but they were not mass produced, and only tried LIBs in PHEV and pure electric versions in the late 2000s in response to Tesla (Colias, 2025). The responses of incumbent firms and their governments, where incumbent automakers started down the road to electric vehicles but stopped short of commercializing battery technology for electric vehicles—until Tesla launched its Roadster in early 2008—created a window of opportunity for latecomers.

BYD and CATL leveraged this window of opportunity and drove the broader shift to EVs in China, together with government support. As a result, the Chinese automotive industry was fundamentally reshaped. Chinese brands dominate in EVs (Fig. 2), and as the share of EVs in total passenger car sales increased from 3 % in 2017 to 40 % in 2024, the share of local Chinese brands in total vehicles sales of the top 10 brands in China increased from 18 % in 2017 to 63 % by 2024, as shown in Table 2. Foreign brands like GM, Ford and Nissan dropped out of the top ten, while Chinese automakers like BYD, Changan, Chery, and GAC entered the ranking and others like Geely and SAIC improved their positions. The majority of EVs in China are supplied by Chinese automakers and predominantly BYD (Fig. 2).

In parallel, CATL and BYD caught up with the global leaders in battery technology from South Korea and Japan. By 2024, CATL led

global EV battery production with a 37.9 % market share, followed by BYD at 17.2 %, significantly ahead of former leaders from South Korea (LG Energy Solutions: 10.8 %; SK On: 4.4 %; Samsung SDI: 3.3 %) and Japan (Panasonic: 3.9 %), measured as a share of globally installed batteries in EVs.¹² BYD produces batteries almost exclusively for its own vehicles. CATL sells batteries to the largest automotive firms in the world inside and outside China, including Tesla, VW, Stellantis, BMW, Mercedes-Benz, Geely. CATL has a much larger global market share than the next biggest Chinese EV battery firms, as Fig. 3 shows.

4. Sectoral innovation system in batteries for consumer electronics

The existence in China of LIB knowledge, manufacturing experience, and a local supply chain supply refining minerals and producing batteries for consumer electronics made it a fertile ground for Chinese firms to emerge that would invest in applying LIB know-how to vehicles. Between the late 1990s and the mid-2000s, China became the largest production hub for consumer electronics products in the world (Sturgeon & Kawakami, 2010; Yeung, 2022). In mobile handsets alone, production in China increased from 2.3 % of global production in 1998 to 37.2 % in 2005. Already in the 1990s, major electronics producers from Japan, South Korea, the US and Taiwan had shifted more and more manufacturing to China as the economy was opened because of its relatively low cost, abundant labour supply. This process accelerated in the 2000s, after China entered the WTO, and coincided with the rise of cell phones and later smartphones, making China the global production hub for electronics.

As part of this process, global battery production became concentrated in China. By 2003/2004, mainland China accounted for 40 % of the global production of nickel-metal hydride batteries, 30 % of sealed lead-acid batteries and, most importantly, 29 % of LIBs – the type of battery that today powers electric vehicles (Patil, 2008). Initially, South Korean and Japanese firms dominated production in China, but local Chinese firms like BYD, ATL (the predecessor of CATL), Shenzhen BAK, Shenzhen B&K, and Tianjin Lishen quickly caught up. Already by 2003, four Chinese firms featured among the global top 10 LIB producers (Ibid: 23). The US and Japan held the major patents in LIBs between the 1970s and 1990s, but this had changed by the 2010s (Gong & Hansen, 2023). This growth in battery firms catalyzed the emergence of a local supply chain in China for refining raw minerals and producing battery parts (Zhao & Luethje, 2024).

The fundamental parts of battery cells – cathodes, anodes, electrolytes and separators – are the same between LIBs for consumer electronics and EVs. Many of the manufacturing processes are similar, and the materials come from the same global mining and refining supply chains. Hence, the relevant materials and component firms already existed in China, as they supplied LIB materials to companies in the large industrial clusters producing electronic and semiconductor products in the Southeast coastal provinces (Luethje et al., 2022). CATL and BYD initially relied on this supply chain before investing in backward integration into battery materials to control innovation, and later investing in mining and refining (by taking shares in existing companies) to control their production costs (Li, 2015). The LIB materials companies also improved over time, through supplying the growing number of battery companies, allowing them to catch-up with the leading Japanese firms in high-performing LIB materials (Wang et al., 2022). This local supply chain also made it easier for new EV battery firms to enter the industry, and some of the LIB material producers also invested downstream in battery cell and system manufacturing.

Thus, China had sectoral innovation systems in batteries for consumer electronics and in ICE automobiles by the early 2000s. But it was

BYD that made the first effort to leverage its knowledge and manufacturing experience in batteries for consumer electronics to move into batteries for electric vehicles. As shown in Section 6, BYD brought its manufacturing knowledge and experience as a supplier of LIBs and electronics products since 1997 and its manufacturing knowledge and experience in ICE automobile production since 2003 to its production of EV batteries and electric vehicles, which it started in 2007. This is how BYD was able to produce at low costs but high quality. CATL, at the time still called ATL, established an EV battery R&D centre in 2006, and invested in mass production in 2008. Without experience in automotive production, it was dependent on finding automakers as partners.

5. Interaction between Chinese firm-level strategies and government industrial policies

BYD and CATL not only benefited from government policies but also actively shaped them. This section shows that the industrial policies that were key to the success of China's EV transition emerged after the two central new energy firms, BYD (both EVs and batteries) and CATL (EV batteries), had made their bets on lithium-ion EV batteries. It provides a timeline of the key strategic events at BYD and CATL based on their business strategies and firm-level technological effort, as well as the key plans and programs of the central government supporting the transition to EVs, summarized in Fig. 4. BYD and CATL's investments came first and interacted with the central government's goals of indigenizing innovation. The result was government industrial policies that supported a general shift to EVs using LIBs based on initially subsidized consumer purchases to create a market and subsidies for Chinese automakers and battery producers linked to compulsions to increase the performance of LIBs. We argue that without the early investments by BYD and CATL, which in turn hinged on China's large consumer electronics industry and the local supply chain in LIBs that served it, the government's industrial policies would likely not have materialized or been so successful.

Central government support for EVs began in 2001 when the Ministry of Science and Technology (MoST) announced the objective of 'zero emissions' for the 2008 Beijing Olympics and launched the 'Major Science and Technology Special Project for Electric Vehicles'. Wan Gang, a former Audi engineer with a PhD in automotive engineering from Germany, oversaw the EV program since 2001 and became Minister of Science and Technology in 2007 (Feng & Li, 2019; Gong & Hansen, 2023). Wan Gang specifically argued that Chinese firms would not be able to outcompete foreign firms in ICE vehicles and thus should 'leap' over them by commercializing technologies for new energy vehicles, the definition of which was kept technology-agnostic, covering non-plug-in hybrid EVs, fuel-cell EVs, plug-in hybrid EVs (PHEVs) and pure battery EVs. It was this realization, as well as the issue of pollution from gasoline cars and the goal of less dependence on foreign oil imports, that triggered government industrial policy to support and advance new energy vehicles. Through the 2000s, policy support came in the form of public R&D funding the technologies underpinning new energy vehicles (Feng & Li, 2019; Tillemann, 2015).

BYD entered the automobile industry in 2003 with an eye to create battery electric vehicles, which will be discussed in more detail in the next section. BYD did not receive any central government funds for R&D until 2007 (Shen et al., 2016). It also went against the central government's recommendation in the R&D program for new energy vehicles from 2001, which urged local producers to adopt existing mature battery technologies rather than aiming for new technologies, when it decided to invest in lithium iron phosphate (LFP) battery chemistry for EV LIBs (Feng & Li, 2019; Shen et al., 2016). When BYD gradually revealed its technological breakthroughs and its PHEV and pure EV models from 2008, government departments adjusted their policies to provide policy support. MoST started to more actively support research on the large-scale production of LFP battery cells and cathode materials (Kamikawa & Brummer, 2024: 8). BYD was listed as the 'vanguard' of

¹² Source: SNE Research, <https://www.sneresearch.com/en/insight/release-view/371/>.

the domestic development of EVs and included in various governmental plans (Shen et al., 2016: 498). A further example is that the central government intervened to stop the Beijing municipal government from discriminating against BYD in favor of firms headquartered in Beijing. After BYD launched its PHEV model F3DM in 2009 and its PHEV model Qin in 2013, most of the automakers in China producing non-plug-in hybrid EVs turned to PHEVs instead, and the government excluded non-plug-in hybrid EVs from eligibility for subsidies in 2017 (Feng & Li, 2019).

Only in 2009 did the central government take a more active approach of market formation. It introduced subsidies for purchases of new energy vehicles by the public sector in selected cities, and required that some cities and public sector organizations have a certain percentage of new energy vehicles in their vehicle fleet (EESI, 2018; Feng & Li, 2019; Gong & Hansen, 2023). These purchase subsidies were extended to private consumers in 2010 in six cities, which coincided with announcements by BYD that it would not launch new battery EV models until the government gave more support to developing the consumer market (The Wall Street Journal, 2009). The consumer subsidies started in Shanghai, Changchun, Shenzhen, Hangzhou, Hefei and Beijing, cities with the headquarters of established automakers. The subsidy covered 40–60 % of the cost of EVs, with higher subsidies for pure EVs compared to PHEVs. In 2013, consumer purchase subsidies were expanded to a total of 88 cities (Altenburg et al., 2017). Only once the purchase subsidies were expanded in 2013 did EV sales started to grow considerably (Fig. 5).

In 2016, the subsidy program for consumer purchases was extended nationwide, with the amount of the subsidy based on the vehicle's driving range per charge. The extension of the subsidy program occurred simultaneously with BYD's launch of several new PHEV and pure EV models as well as other carmakers' EV models based on CATL batteries. From 2015, the central government also increased subsidies for new energy vehicle infrastructure (e.g. charging stations, battery swapping systems), which had been very limited previously (Gong & Hansen, 2023). The exemption on licence plate control, discounted/free electric charging and preferential bus-lane access also contributed to higher EV acceptance and purchases (Wang et al., 2022). As a result, EV sales continued to accelerate (see Fig. 5).

Expected increased EV sales as a result of the subsidy programs, and thus increased demand for EV batteries, led the central government to ensure that Chinese battery firms benefited from this demand. In 2015 it released the 'Automotive Power Battery Industry Specification', commonly known as the Battery Whitelist, which stated that only NEVs that were equipped with batteries that met the conditions specified in the document were eligible to receive subsidies (Gong & Hansen, 2023: 11). One condition was that only batteries made by local firms were included in the list. The Battery Whitelist, which lasted from 2016 to 2019, was a key industrial policy. It was classic 'infant industry protection' against the globally leading Japanese and South Korean battery firms that had been operating in China up until then, and it provided secure and protected market demand and thus scale economies required for Chinese battery firms to engage in learning-by-doing. However, the initial subsidy regime from 2009 already had the condition that the battery, as well as the electric motor and the inverter, had to be made by a majority Chinese-owned entity (Kamikawa & Brummer, 2024: 9). Already in mid-2018, the central government changed the Battery Whitelist to allow three Chinese joint ventures with the South Korean firms Samsung, LG Chem and SK Innovation. Their inclusion sent a strong signal of relaxing protectionism and promoting stronger market competition, which was followed by the end of the Whitelist in 2019 and liberalization of automotive foreign direct investment (Wang et al., 2022).

Another important condition in the Battery Whitelist related to the technical requirements for batteries. It gradually tightened the technical requirements from 2016, so that batteries had to reach a maximum speed and minimum driving range, and thus required higher battery density and increased vehicle energy efficiency (Wang et al., 2022). As a result, most Chinese battery firms focused on nickel-cobalt based battery chemistries, as they had high energy density. But it also encouraged BYD and CATL to improve the speed and range of their LFP batteries. These firms announced their new EV batteries based on cell-to-pack design in 2019 and 2020, respectively.

We argue that EV sales on the Chinese market only took off after 2020, as shown in Fig. 5, because Chinese automakers other than BYD did not make major investments in PHEV and pure EV models until the late 2010s, when government policies combined with the efforts of BYD and CATL had succeeded in creating large consumer market demand for EVs and lower-cost LIBs with an improved range and charging performance. As shown in the next section, these conditions made it easier for Chinese ICE automakers to shift into EV production. As EV subsidies were reduced annually, ending in 2022, automakers producing EVs absorbed the reduction through the lower costs of batteries (Luethje et al., 2022: 15). The result was that by 2019 battery prices fell by 20–40 %, making the average battery pack price in China the world's lowest.

The moves of Chinese automakers to offer EV models were further catalyzed by the government's announcement of the New Energy Vehicle Development Plan 2021–2035 in October 2020 which stipulates that by 2035 pure EVs should become the "mainstream" of new vehicle sales in China (General Office of the State Council, 2020). Additionally, the shift to EVs was spurred by the introduction of the NEV mandate that set targets for automakers in China to transition to new energy vehicles. It started at 10 % of the conventional passenger vehicle market in 2019 and 12 % in 2020 for any carmaker with over 30,000 vehicles manufactured locally or imported yearly in China. Automakers had to buy credits from other automakers to avoid penalties if they failed to meet the targets. Both BYD and Tesla generate in excess of \$1bn per year through this credit trading system (Shou & Li, 2024).

6. BYD and CATL's catch-up compared to other Chinese EV and EV battery firms

All Chinese firms benefited from the same local supporting nexus and government industrial policies, but BYD and CATL outperformed all other Chinese EV firms and EV battery producers by a significant margin. In this section, we explain why, drawing on the technological catch-up matrix. We compare BYD to the other Chinese firms among the top 10 EV producers in China (i.e. eight other firms, excluding Tesla), and CATL to the five other Chinese firms in the global top 10 EV battery firms (see Figs. 2 and 3). Due to limited space, the text focuses on the stories of BYD and then CATL, while Tables 3 and 4 summarize the relevant information of the comparator Chinese EV and EV battery firms. The online appendix provides a complete list of sources on the comparator Chinese firms that we used to produce our analysis in Tables 3 and 4.

This comparative exercise shows that what sets BYD and CATL apart from their Chinese peers was their existing knowledge and capabilities combined with their firm-level effort to acquire new knowledge and capabilities. Specifically, BYD and CATL moved into EV batteries earlier, started with greater know-how and organizational capabilities in battery manufacturing, and formed earlier and more intensive partnerships which linked them to the foreign technological frontier in batteries and automobiles.

Table 3

Key Characteristics of the top Chinese EV producers (2024) based on the technology catch-up matrix.

Chinese EV producers	Background	Year first EV produced with LIB	Access to capital for investment	Automobile knowledge and capabilities	EV battery knowledge and capabilities	Market demand	Local supporting nexus
BYD	Consumer electronics batteries (1995); ICE cars (2003)	2008	Other businesses: BYD Electronics; IPO in Hong Kong; Warren Buffett investment	ICE automaker since 2003; Daimler collaboration (2010); Hiring of experienced German designer as Head of Design (2016)	Consumer battery production (since 1990s); EV battery R&D efforts (since mid-2000s); Internal EV battery cell production for all its vehicles since 2008	Local government procurement (Shenzhen); subsidized infrastructure & consumer subsidies for EVs nationally	Lithium battery materials local supply chain; Vertical integration in auto parts and EV raw materials
Geely	ICE cars (1997)	2015	Incumbent business; IPO in China and US	ICE automaker since 1997; acquired Volvo in 2010; Geely's founder has a 9.69 % equity stake in Mercedes-Benz since 2018	Mainly external sourcing; smaller investments in battery firms in the mid-2010s; JVs with CATL (2018) and LG Chem (2019) for joint cell production; in-house R&D for own batteries; start of internal battery cell production in 2024	EV market in China already existed when it entered; started overseas EV sales early in 2017 through its subsidiary Polestar; more than 1/3 of total production exported in 2024	EV battery supply chain in China (CATL major supplier, but also uses several other suppliers); Chinese software firms (e.g. partnership with Baidu for software development)
SAIC-GM-Wuling	ICE cars (2002)	2017	Incumbent businesses of the JV partners; both SAIC and Wuling state-owned	Both SAIC and Wuling made ICE cars since the mid-20th century; SAIC-GM-Wuling founded in 2002	Battery cells fully externally sourced; JV with CATL (2017)	Market already existed; Exploited demand for city mini-car	EV battery supply chain in China (mainly sources from CATL, Gotion, Sunwoda)
Changan	ICE cars (1959)	2012	Incumbent business; state-owned; external co-investors in the two largest EV brands (Deepal and Avatr)	ICE automaker since 1959, with volume production since the 1980s	Mainly external sourcing; JV with CATL (2018); internal battery R&D; internal battery cell production announced; aims for internal production of solid-state batteries by 2027	Its first EV model, the EV30, was sold in Beijing as a taxi (local government demonstration project).	EV battery supply chain in China (LG Chem for its first model; Chinese suppliers today); partnership with CATL for its internal battery factory; Chinese software firms
Li Auto	E-commerce (2002)	2019	Meituan and ByteDance investments; IPO in US and Hong Kong in 2020	Bought factory of Lifan, existing auto company; partnerships with suppliers Bosch, BorgWarner, Panasonic, Valeo	Battery cells fully externally sourced	Market already existed; strategic focus on premium SUVs (PHEV)	EV battery supply chain in China (CATL major and initial supplier; recently added SVOLT and Sunwoda)
Chery	ICE cars (1997)	2010	Incumbent business	ICE automaker since 1997	Battery cells fully externally sourced; internal battery cell R&D; aims for internal production of solid-state batteries by 2027	Initially focused on small city car EV models; only later developed broader product range	Sourced batteries from China BAK for its first model, and later mainly from CATL
Seres	Auto parts (1986); ICE cars (2005)	2019 for passenger EVs (but commercial EVs (vans) since 2015)	Incumbent business; shareholders include local government, Dongfeng (SOE) and private investors; IPO Shanghai 2016	ICE automaker since 2005; set up office in Silicon Valley in 2016 to tap into EV knowledge; acquired Tesla founder Martin Eberhard's EV startup in 2017 and hired him as Chief Innovation Officer 2017-2018	Sources batteries fully externally, mainly from CATL; but proprietary battery cell designs since 2018 (developed in R&D centers with small-batch test production lines in China and California)	Market already existed; initial focus on minivans; recent focus on high-end premium SUVs together with Huawei (AITO brand) (distribution through Huawei stores)	Initial growth (1990s) as parts supplier in motorcycle production cluster in Chongqing; AITO brand vehicles are co-developed with Huawei and distributed through Huawei's stores; Long-term battery supply agreement with CATL
GAC	ICE cars (1997)	2015	Incumbent business; state-owned; external investors for EV business Aion in 2017	ICE automaker since 1997; EV business spun-off in 2017 and fully independent since 2020 as GAC-Aion	Mainly externally sourced; internal battery cell R&D since 2017; internal battery cell production since late 2023	Market already existed	Initial batteries from CATL, but today CALB major supplier; JV with CATL (2018)
Great Wall Motor	ICE cars (1984)	2017	Incumbent business; IPO Hong Kong 2003 and Shanghai 2011	ICE automaker since 1984; JV with BMW for EV Minis in 2018	EV battery R&D in-house since 2012, production since 2016, spun-off as SVOLT in 2018	Market already existed; focus on pick-up trucks and SUV niche	Vertically integrated in auto parts; spun off internal battery production in 2018; still sources from that spin-off, but also from external suppliers like CATL

Source: Authors' elaboration based on sources detailed in the online appendix.

Table 4

Key Characteristics of the top Chinese EV battery cell producers (2024) based on the technology catch-up matrix.

EV battery cell producers	Background	Start EV battery cell mass production	Access to capital for investment	EV battery knowledge	EV battery production capabilities	Initial market demand	Local supporting nexus
CATL	Batteries for consumer electronics (founded as ATL in 1999); spun off from ATL in 2011	2006 (as ATL)	Private investment from founders and other Chinese private investment; IPO in Shenzhen in 2018	Origins in Japanese electronics firm TDK; Internal R&D, started under ATL; Recruitment of expert from GM to head R&D	Learning-by-doing; partnership with BWM (2011)	Electric buses initially; Partnership with BMW-Brilliance; BYD stopped selling batteries to other EV producers	ATL's initial factory was in Guangdong, accessing local customers and the battery supply chain; own vertical integration investments in materials and mining since 2020
BYD CALB	See Table 3 Incubated by the China Air-to-Air Missile Research Institute, under the state-owned Aviation Industry Corporation of China (AVIC) (2007)	2006 2009 (initial focus on commercial vehicles)	See Table 3 Subsidiary of AVIC; taken over successively by Jintan, Changzhou District government since 2015 with repeated capital injections; IPO in Hong Kong in 2022	Started with focus on LFP batteries for commercial vehicles, mainly buses; more focus on nickel-based since 2018	Learning-by-doing (focus on LFP since inception)	Initial focus on commercial vehicles; first PHEV customer: local government demonstration project in 2010; today supplies GAC, Changan, SAIC-GM-Wuling, Leapmotor and XPeng.	Link to AVIC and military industry; proximity to East China electronics cluster around Shanghai; Jiangsu province has a large concentration of power battery firms
Gotion High-Tech	Founded as EV battery producer (2006)	2009 (first for electric buses)	Founded by real estate entrepreneur with links to local Hefei government; IPO in Shenzhen in 2015; VW investment in 2020	Internal R&D; started with focus on LFP	Learning-by-doing (focus on LFP since inception); partnership with VW (2020)	Supplying commercial vehicles under the 2009 government EV subsidy program; first OEM customer JAC in 2011; buses still major focus, but also passenger EV OEMs, incl. VW	Link to local OEM JAC; Proximity to East China electronics cluster with additional plants there; Huawei is a major customer of batteries for base stations
EVE Energy	Lithium primary (non-rechargeable) batteries for consumer electronics (2001)	2015	Incumbent business; IPO in Shenzhen in 2009	Internal R&D; JV with SK Innovation in 2018	Learning-by-doing (initial focus on lithium primary batteries; entered LIB in 2011, and EV batteries in 2015); JV with SK Innovation in 2018	Commercial vehicles; Started supplying big automakers through SK Innovation from 2018	Based in Guangdong with access to strong local supply chain
Sunwoda	LIB for consumer electronics (1997)	2018	Incumbent business; three rounds of financing between 2018 and 2024	Internal R&D; focus mainly on nickel-based	Learning-by-doing; partnerships with foreign firms between 2008 and 2015	Producing for Li Auto, Geely and other EV producers looking for fast-charging battery tech	Based in Guangdong with access to strong local supply chain

Source: Authors' elaboration based on sources detailed in the online appendix.

6.1. BYD compared to other Chinese EV producers

We explain BYD's dominance in EVs as the result of its unique combination of mastering knowledge and developing organizational capabilities in both the battery industry and the automotive industry over a long period of time. It started making LIBs for consumer electronics in the 1990s and making automobiles in the early 2000s, moving into LIBs for EVs in the mid-2000s. BYD is the only EV producer of the top Chinese EV producers listed in Table 3 that has a history in LIB manufacturing.

BYD started as an electronics firm in 1995 based in Shenzhen. The founder Wang Chuanfu had a background in battery chemistry and worked at a government research institute, before starting to produce nickel-based batteries for Chinese toy manufacturers exporting to Europe. Wang aimed to improve the quality of batteries, and studied new technologies around LIBs that the Japanese firm Sanyo was using. Wang's small R&D team studied the patents and reverse-engineered Sanyo's batteries, improving its chemistry formula and filing its own patents. He and his team also visited Japan in 1997 to see the automated production lines of Japanese battery firms. However, the Japanese firms would not share their technology or sell production equipment. Wang invested in developing his own production process and equipment (Qin & Xiong, 2024: 29–39). He also changed the battery production process

to be more labor-intensive, instead of the robots used in Japanese factories, and as a result, achieved parity in quality with Japanese firms but with lower production costs through industrial engineering process innovations and cheaper female migrant labor (Li, 2015). With financing raised through its IPO in 2002 on the Hong Kong stock exchange, BYD invested in producing other components for mobile handsets, so as to move into phone assembly and become a contract manufacturer like Foxconn (Huckman & McCormack, 2009).

BYD began filing its own patents in 1999, but in the early 2000s, two leading Japanese battery firms Sanyo and Sony filed patent infringement cases in US courts against BYD's batteries for mobile phones and laptops (Qin & Xiong, 2024: 64–70). The Japanese actions were in response to increasing competition from BYD. By 2002, BYD was the largest Chinese supplier of rechargeable batteries to cell phone manufacturers, and the fourth largest supplier globally, with Motorola and Nokia as its key customers.

BYD Electronics won the patent infringement cases, and by 2007, it became the world's largest LIB manufacturer, supplying Apple and Samsung. BYD gained important manufacturing experience from supplying Motorola and Nokia and then Apple and Samsung, building its knowledge and organizational capabilities. BYD had over 100 patents by 2002. Supplying these lead firms also helped BYD Electronics to stay in step with advancing technology (Qin & Xiong, 2024: 52–57).

Wang took note of the potential of LIBs for electric vehicles already in the late 1990s (Huckman & McCormack, 2009; Sina, 2008; Wang, 2024). Using the IPO proceeds, Wang entered the automotive industry in 2003 by acquiring the assets of the bankrupt state-owned automotive company Qinchuan Auto and its government permit (Shen et al., 2016; Sina, 2008; Wang & Kimble, 2010: 79). BYD Auto produced the F3, an ICE car similar to the Toyota Corolla, based on reverse engineering the non-patented technical tasks and creatively changing other parts to get around patents (Qin & Xiong, 2024: 101). In 2009, the F3 became a best-selling model in China. Its strategy was to achieve parity in quality at a much lower price, through vertical integration (producing most components in-house) and using the same labor-intensive manufacturing model that it had in consumer electronics (Gong & Zhu, 2015).

Wang started an EV R&D project at the same time as developing capabilities in producing ICE vehicles. BYD showcased its fully electric car concept with LIBs in late 2006 and started its industrialization program for lithium-iron phosphate (LFP) EV batteries in 2007. Wang chose LFP as the cathode material for batteries based on findings from a 2005 study by MIT researchers (Shen et al., 2016: 498). This technology choice was driven by considerations around access to patents (see Kamikawa & Brummer, 2024), and the companies' experience with this technology from the consumer electronics sector. Additionally, Chinese EV customers initially were more sensitive to price and less sensitive to the driving range of the battery, making LFP batteries a better choice because they were cheaper to produce but had a lower driving range compared to nickel-cobalt based batteries (Frith et al., 2023). As discussed in Section 5, BYD's choice of battery technology was not in line with the central government's view at the time.

Based on its internally produced batteries, BYD launched its first new energy vehicle in late 2008, the F3DM plug-in hybrid. It could produce pure EVs but started with PHEV due to limited charging facilities in the country; however, its model had a higher range than other hybrids on the market at that time (Gong & Zhu, 2015). BYD first promoted its F3DM to local governments and companies, as they could build the required charging infrastructure and because of the relatively high price of the F3DM.

BYD's first pure EV model, the E6, was launched in 2009, for the Shenzhen municipality for use as taxis (Feng, 2010: 218). Thus, BYD set out to produce pure EVs around the same time as Tesla. It expanded into buses and taxis in 2011, using a leasing payment system, and benefiting from central and local government subsidies. With this market demand, BYD started to mass produce EV batteries and also acquired a stake in a lithium mining and refinement company to secure raw materials, vertically integrating most aspects of the local supporting nexus (Gong & Zhu, 2015; Shen et al., 2016; Zhao & Luethje, 2024).

Warren Buffet saw the potential of BYD and acquired a 10 % stake in its stock in 2008, which strongly supported the company in its early years. Buffet's investment in BYD provided capital for investment and increased the confidence of other investors. In addition, it attracted the interest of German incumbent automaker Daimler (Altenburg et al., 2017). BYD entered a joint venture with Daimler in 2010 to develop EVs for the high end segment, comparable to Tesla, and launched the Denza brand in 2014. Denza adopted the battery system, powertrain technology and control system of BYD's E6 model and uses the front and rear suspension of Mercedes-Benz's series B. Daimler was responsible for vehicle design, molds and safety – areas where BYD wanted to learn (Shen et al., 2016: 498; Zhao & Luethje, 2024: 21–22). Wang later hired the German design director of Audi in 2016 to lead BYD's design team (Qin & Xiong, 2024: 197).

Nevertheless, BYD Auto struggled financially until 2020, as its sales were low, and revenue from BYD Electronics was important to keeping it afloat (Qin & Xiong, 2024: 185–221). It needed to improve the battery range of its pure EVs, which Tesla had already achieved using nickel-based lithium battery chemistry and not LFP. BYD's innovation efforts with LFP eventually resulted in the blade battery in 2020, with

higher energy density and thus higher range that could compete with nickel-based battery chemistries in terms of cost-performance ratio (Frith et al., 2023; Zhao & Luethje, 2024).

The launch of its blade battery and the Han EV model were key turning points for BYD. After that, its investments came together to form a great synergy: transfer of manufacturing capabilities from BYD Electronics, heavy investments in R&D, and vertical integration where it treated its subsidiaries as suppliers to increase competition, as well as some fortune when global nickel prices increased which favoured LFP batteries (Qin & Xiong, 2024: 226). These factors led to high performance and lower costs than its competitor Chinese EV producers.

When we analyzed the capabilities and market performance of the other top Chinese EV producers, we see that most have experience producing ICE cars for at least two decades. Notably, none of the Chinese EV start-ups by founders with a background in software and e-commerce, such as NIO, Leapmotor and XPeng, were among the top 10 EV producers in China as of 2024. These EV start-ups had strengths in digital technologies, but they lacked knowledge and organizational capabilities in producing both automobiles and LIBs, which they either had to build from scratch or outsource.

An EV is still a car, and despite the EV start-ups loading it with high-tech features, driving range and charging time related to the battery technology and the integration of the various components for a good driving experience still matter a lot (Alochet et al., 2023). Among the many EV start-ups in China, only Li Auto made it into the top 10 list. Unlike NIO, which fully outsourced automobile assembly until very recently, Li Auto purchased a factory from an auto company immediately after being founded in 2015, and relied on technology collaboration with large automotive suppliers to bridge the capabilities gap. However, the tacit knowledge and organizational capabilities required for producing cars at scale take time to build.

6.2. CATL compared to other Chinese EV battery cell producers

BYD is the only producer of EVs that has made its batteries in-house since inception. All the other Chinese automakers externally sourced their EV batteries, mainly from CATL, at least until very recently. This reliance of EV producers gave CATL a monopoly-like position in the Chinese market, and put other Chinese battery producers in a position of trying to catch-up with CATL. We ascribe CATL's market leadership to its early entry into EV battery production and its early efforts at accessing and mastering automotive knowledge. The other Chinese battery firms either started producing EV battery cells much later or did not tap into automotive knowledge to the extent that CATL did and as early as CATL did (see Table 4).

CATL emerged in 2011 as a spin-off of Amperex Technology (ATL), which has a trajectory similar to BYD in that it started out making LIBs for electronic products. Robin Zeng founded ATL in Hong Kong in 1999 and made batteries for mobile electronics such as MP3 and DVD players based on a lithium polymer patent acquired from Bell Labs in the US (Sanderson, 2022). Robin Zeng had worked for a subsidiary of the Japanese electronics firm TDK in Hong Kong before he was convinced by his TDK managers to set up ATL, supplying batteries for Apple and Samsung smartphones. The link with TDK helped with access to technology and knowledge. When the major US shareholder in ATL pulled out in 2005, TDK acquired it (36Kr 2022).

Zeng decided in 2006 to establish an internal EV battery development department, which initially focused on batteries for electric buses and succeeded in supplying batteries for the electric buses during the 2008 Beijing Olympics (Wards Auto, 2018). In 2008, Zeng invested in a large-scale EV battery cell factory in Ningde, which became fully operational by 2012 (Hall, 2014: 62). But the real turning point was when BMW approached Zeng in 2010 to supply the batteries for the pure EV model of BMW's joint venture in China, BMW-Brilliance.

BMW, an early mover in EVs compared to its German peers, wanted to source batteries from BYD for its electric vehicles, but BYD declined,

as it decided not to provide its EV batteries externally (Quartz, 2019). BMW then approached ATL because of the company's reputation as an Apple supplier and one of the largest battery manufacturers for consumer electronics (Hall, 2014). ATL's owner TDK was not very interested in a major shift towards LIBs for EVs, so Zeng decided, jointly with co-investors, to buy out 85 % of ATL's EV battery business in 2011, founding CATL (36Kr 2022).¹³ Buying out TDK also helped to fulfil the new energy vehicle subsidy requirement of Chinese ownership of battery technology. The agreement with BMW was finalized and the resulting vehicle, the Zinoro 1E, was launched in December 2013.

CATL's collaboration with BMW was essential for developing organizational capabilities to supply the automotive industry and for accessing knowledge related to automotive LIBs, especially given that the Zinoro 1E was the first premium pure EV model launched in China (BloombergNEF, 2018; Quartz, 2019; Sanderson, 2022: 42; Wired, 2022). CATL made the entire battery pack but worked closely with BMW to design it in a way that it fit the requirements of the overall vehicle design (Hall, 2014: 69). Throughout the collaboration, BMW had staff at CATL to follow the battery manufacturing process. The partnership with BMW also lifted CATL's profile and attracted the interest of Chinese automakers, resulting in multiple supply deals throughout the 2010s and even joint ventures with various Chinese and Western automakers for battery production (Financial Times, 2018; Wards Auto, 2018).

Equally important as the arrangement with BMW-Brilliance for knowledge acquisition and capability building in EV battery production was Zeng's decision to hire Bob Galyen, the former head of battery development for GM's EV1 model (1996–1999), as CATL's CTO in 2011. Galyen headed CATL's EV battery R&D department until 2019 and built up its passenger EV battery business (Colias, 2025: 33–37; Hall, 2014; Sanderson, 2022).

CATL improved LFP battery energy density through cell-to-pack battery design innovation. This kind of incremental innovation in process and product comes from scale economies and associated learning effects and accumulated tacit knowledge that CATL gained from producing EV batteries since the late 2000s (Wired, 2022). When CATL launched its new cell-to-pack technology in 2019, it changed the EV battery scene not only in China but also globally, as it could outcompete existing NMC batteries on the price-performance ratio (see Frith et al., 2023). It further advanced its Qilin battery over the next years in terms of range and fast charging.

In addition to CATL's own learning efforts, the removal of foreign battery firms from the Chinese market between 2016 and 2018 via the Battery Whitelist played an important role in giving CATL a secure, protected market. As the consumer LIB producer with the strongest knowledge and capabilities in EV battery manufacturing, CATL managed to meet the government's increasing energy density standards linked to the whitelist, unlike most of the other Chinese battery producers. CATL had strong capabilities in LFP and nickel-cobalt-based battery chemistries that were favoured by the Whitelist's energy density criteria and thereby captured significant market share in those years (Gong & Hansen, 2023). CATL also benefited from BYD's decision to stop selling its batteries to other EV producers.

7. Conclusion

This article set out to address significant gaps in explaining the rise of Chinese firms like BYD and CATL to global leadership positions in electric vehicles and EV batteries. The introduction identified two key limitations in current approaches. First, conceptualizations of local firm upgrading in the GVC and GPN frameworks struggle to explain how latecomer firms can improve their position in the global production network hierarchy by becoming lead firms themselves. Second, most

literature on China's EV transition emphasizes government industrial policies, which often has the effect of downplaying the role of firm-level technological effort.

Our analysis addresses these gaps by demonstrating how BYD and CATL achieved technological catch-up and leapfrogging in EVs and EV batteries through systematically building the proprietary knowledge and organizational capabilities required to challenge established industry hierarchies in the automotive industry. Both BYD and CATL built key battery manufacturing capabilities as suppliers to lead firms in the consumer electronics industry, and thus strategic partnerships were important, but equally important were the firms' technological efforts from the beginning. This was illustrated by BYD's internal R&D investments in mastering lithium-ion battery production when it could not access technology from Japanese incumbent firms. The technological catch-up matrix provides a systematic framework for analyzing how firm efforts and government policies interacted across the five dimensions: proprietary knowledge, organizational capabilities, access to patient capital, sufficient market demand, and a local supporting nexus.

BYD and CATL used transnational networks extensively to access and adapt foreign knowledge and build organizational capabilities. Government industrial policies cannot mandate such firm-level efforts, but they can provide the incentive structure under which they are more likely to occur and to be successful. Industrial policies were particularly important in providing the necessary market demand through the various market incentives and subsidies for NEVs, as well as via granting temporary market protection for local EV battery firms. China's overall state-directed approach towards industrialization is one of the reasons why there is such a strong local supporting nexus, with growing automobile production and strong consumer electronics and battery production since the 1990s that firms could draw on for their starting capabilities and inputs, as well as adequate access to capital for manufacturing investment. This highlights how technological catch-up does not occur in a vacuum, but critically depends on the local conditions, which economic geographers call regional assets and innovation economists call sectoral innovation systems. Nevertheless, the role of firm effort was crucial in these dimensions too. To access the necessary capital for fixed investment, firms engaged in cross-subsidizing from their other business units, especially in the case of BYD, as well as leveraging foreign investment through IPOs and other means. A focus on the local supporting nexus alone cannot explain why several other Chinese battery firms did not enter EV batteries early on.

Our analysis of BYD and CATL compared to other Chinese EV and battery firms reveals the centrality of firm-level technological effort alongside government support. While all Chinese firms operated within the same policy environment and benefited from similar industrial policies, BYD and CATL outperformed their competitors by substantial margins because of their earlier entry, superior starting capabilities in battery technology, and more intensive efforts to access foreign knowledge through partnerships and talent recruitment. Furthermore, BYD and CATL's early investments in lithium-ion battery technology for EVs, beginning in the mid-2000s, preceded and actively shaped government industrial policies rather than simply responding to them. The synergies between industrial policies and technological effort of local firms were key, and the ways in which they drove each other were dynamic and multi-dimensional.

Our findings contribute to broader debates about industrial transformation and development by showing how windows of opportunity – in this case, incumbent automakers' delayed response to EV technology – can enable reversals in global industry leadership (see Lema et al., 2020). The Chinese case demonstrates that such reversals require alignment between firm-level technological effort, supportive institutional environments, and favorable sectoral preconditions. For GVC/GPN literature to be able to engage meaningfully in such debates, we suggest moving beyond the supplier upgrading focus of traditional GVC literature and the lack of theorizing firm-specific initiatives of capability building in the GPN strategic coupling approach towards a

¹³ CATL bought out the remaining 15 % from TDK in 2015, and Zeng fully left ATL in 2017 to exclusively focus on CATL (Financial Times, 2018).

stronger focus on firm-level technological effort, using available tools from the economics of innovation literature. Our technological catch-up matrix tries to synthesize these tools to provide a framework for understanding how firms in emerging economies can transition from subordinated positions in global networks to industry leadership through strategic technological effort.

CRedit authorship contribution statement

Lindsay Whitfield: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tobias Wuttke:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

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